

Testing pure components

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Least privilege and tool scopes

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Configuration and secrets

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Requirements and service boundaries

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Testing pure components. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Testing pure components. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Passing unit cases for a pure function does not establish integration permissions or model-dependent behaviour. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled testing pure components evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of testing pure components. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Testing pure components

Test a room formatter alone. A pure function returns "Room " plus a supplied positive integer.

Accepted inputs are positive integers. Expected outputs: $7 \rightarrow \text{"Room 7"}$; $12 \rightarrow \text{"Room 12"}$.

Testing pure components

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Call with 7

The prefix and supplied numeral produce Room 7.



2. Compare exact text

The expected text includes one space before 7.



3. Call with 12

The same fixed procedure produces Room 12.

CHANGE ONE THING

Supply zero instead of 7.

Explain the changed result using the matching lesson.

Predict the formatter outputs for 7 and 12, including spacing. Why does input 0 need rejection under this contract rather than an ordinary formatted response?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Test a room formatter alone. A pure function returns "Room " plus a supplied positive integer.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The same fixed procedure produces Room 12.

1. Call with 12
2. Compare exact text
3. Call with 7

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Call with 12 The same fixed procedure produces Room 12. A pure component depends on its accepted input, not a clock, network or model. Isolating it makes failures easier to locate.

Practice 2: explain the deciding evidence

Changed-case practice: Supply zero instead of 7. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Reject the input under the stated positive-integer contract.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject the input under the stated positive-integer contract. Zero has the right type but violates the value requirement.

Repair a missing building block

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the mechanism and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied. Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the mechanism and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the mechanism and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

